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Évaluation et validation des différentes méthodes pour mesurer la posture de la tête

Al-Yassary, Mustafa

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Section de médecine Dentaire
Département de réhabilitation oro-faciale Clinique
universitaire de médecine dentaire
Division d'Orthodontie

Thèse préparée sous la direction du Professeur Stavros KILIARIDIS et Gregory ANTONARAKIS

" Évaluation et validation des différentes méthodes pour mesurer la posture de la tête "

Thèse
présentée à la Faculté de Médecine
de l'Université de Genève
pour obtenir le grade de Docteur en médecine
par

Mustafa Al-Yassary
D'
Aigle

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INTRODUCTION

LA POSTURE NATURELLE DE LA TÊTE (NHP)

La posture naturelle de la tête (NHP) est définie quand "un homme est debout et que son axe visuel est horizontal, sa tête est dans une position naturelle"[1] L'évaluation de la NHP est réalisée sur trois axes, à savoir le Pitch, le Roll et le Yaw. Le « pitch » est l'angle entre le plan horizontal et le plan de Francfort. Le « Roll » est l'angle entre le plan horizontal et le plan bipupillaire. Enfin, le « Yaw » est l'angle entre le processus acromial et le plan sagittal médian[2]. (fig. 1)

La posture de la tête joue un rôle important en orthodontie, car elle permet d'évaluer plusieurs aspects essentiels. Tout d'abord, un diagnostic précis de la posture de la tête aide à mieux comprendre la relation entre le maxillaire et la mandibule sur une radiographie céphalométrique, ainsi que les tissus mous sur une photographie du profil, ce qui est fondamental pour établir un plan de traitement adapté. De plus, un diagnostic précis permet d'élaborer un plan de traitement plus efficace. Il a été observé que des postures anormales de la tête peuvent être liées à des perturbations de la croissance des maxillaires[3], des malocclusions[4] et ainsi une disharmonie entre les deux mâchoires. En suivant la posture de la tête de manière standardisée, l'orthodontiste peut mieux accompagner le patient tout au long du traitement, garantissant ainsi un résultat fidèle aux attentes initiales.

En pratique, la NHP est mesurée de deux manières différentes : la position auto-équilibrée et la position auto-guidée. La position auto-équilibrée (self-balanced) est adoptée lorsque la personne regarde un point imaginaire devant elle, à une distance infinie au niveau des yeux, puis la personne va bouger la tête sur un axe à la fois, tout en fixant le point, jusqu'à ce qu'elle trouve la position la plus confortable[3]. La position auto-guidée (mirror-guided head posture) est lorsque la personne se regarde (dans les yeux) dans un miroir[5]. La posture de chasse (hunter position) est une position standard intéressante qui est adoptée lorsque la personne regarde un certain point quel que soit sa position[6]. Mais peu d'études se sont intéressées à cette dernière, et nous n'avons pas d'informations à propos de ses caractéristiques.

Différentes études indiquent que la position de la tête guidée par miroir est la plus stable et reproductible[3, 7] ; cependant, elle est rarement adoptée dans la vie quotidienne. Il est donc préférable, pour simuler plus précisément la position de la tête dans un environnement naturel, et habituel d'analyser la position auto-équilibrée de la tête, qui est adoptée plus fréquemment.

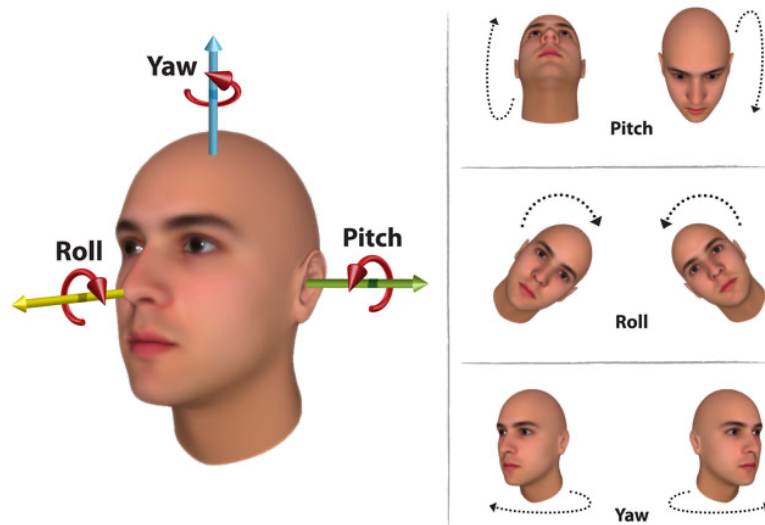


Figure 1 : représentation des 3 axes de mouvement de la tête (Arcoverde Neto et al., 2012)

MÉTHODE POUR MESURER LA NHP

Différentes méthodes existent pour analyser la posture de la tête. La méthode subjective est une interprétation de l'opérateur pour évaluer la NHP. À des fins de recherche, où une évaluation objective de la posture de la tête est recherchée, il existe trois types de méthodes pour évaluer la posture de la tête :

1. ÉQUIPEMENT MONTÉ SUR LA TÊTE : Aussi appelé un « head-mounted equipment », il s'agit de la méthode gold standard pour évaluer la posture de la tête, considérée comme la référence. Un bon exemple est le « The Cervical Range of Motion » (CROM)[8] (fig. 2). Cet appareillage est porté comme des lunettes, il dispose deux inclinomètres permettant de mesurer l'axe du Pitch et Roll. L'inclinomètre est composé d'un boîtier circulaire, graduée (tous les uns degré) et d'une bille métallique. Grâce à la force de pesanteur la petite bille se déplace pour indiquer le degré de positionnement de la tête. L'inclinomètre en face du fond indique le Roll et celui qui est disposé sur le côté du patient indique le Pitch. The Yaw est indiqué par une boussole (graduée tous les deux degrés) au-dessus du patient, des aimants sont placés sur les épaules pour indiquer la référence du nord.

Une lecture erronée peut se produire si le dispositif est mal placé ou si le patient présente une asymétrie faciale. De plus, lors de l'installation du dispositif, le patient peut adopter une position

différente en raison du poids de l'appareil. L'équipement monté sur la tête peut également être utilisé avec des gyroscopes électroniques.

Ces dernières années, les capteurs portables composés d'accéléromètres et de gyroscopes, communément appelés unités de mesure inertielle (IMU : inertial measurement unit). Elles sont devenues de plus en plus populaires pour mesurer le mouvement des articulations. L'IMU permet de mesurer les angles articulaires en utilisant l'orientation dans l'espace collectée à partir de deux capteurs[9, 10]. Les avantages de ce système sont sa facilité d'utilisation, la petite taille et le poids très léger ainsi que sa précision. Un désavantage de ce système est qu'il nécessite un calibrage pour identifier la position de référence.

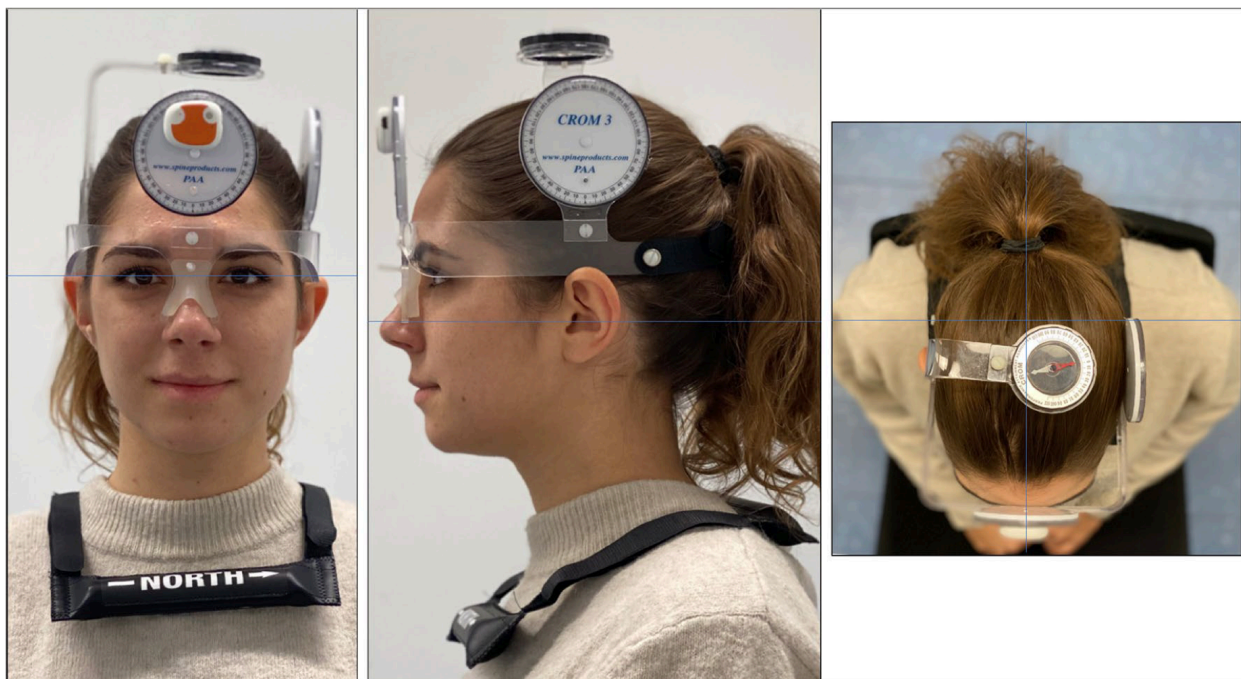


Figure 2 : photographie des 3 axes du Cervical range of motion. On voit aussi un capteur (inertial measurement unit) placé sur l'inclinomètre frontale. (Al-Yassary M. et al 2021)

2. ANALYSES PHOTOGRAPHIQUES BIDIMENSIONNELLES : l'utilisation de photographies est une autre méthode pour évaluer la NHP. C'est simple, rapide, peu coûteux et donne des résultats précis. Contrairement à la méthode précédente, cela ne nécessite pas au patient de porter un équipement encombrant. Cependant, l'axe de la photographie doit être très précis, et il peut être difficile de l'adopter avec des patients en mouvement[11]. De plus il est difficile de mettre en place un système qui peut capturer les 3 axes de manière simultanée ou sur une longue

durée. Le même principe peut être utilisé en radiologie, mais cela expose le patient à des radiations inutiles.

3. ANALYSES TRIDIMENSIONNELLES (3D) : Les scans optiques tridimensionnels de la tête peuvent également être utilisés pour mesurer la NHP (fig. 3). Le patient n'a pas besoin de porter d'équipement et certains systèmes ont une très grande précision. Cette méthode nécessite une configuration complexe, est coûteuse et nécessite une bonne connaissance du logiciel associé. De plus une analyse sur une longue durée est très complexe à réaliser avec ce système[5].

Encore une fois, le même principe peut être utilisé en radiologie, mais cela expose le patient à des radiations encore plus importante.



Figure 3 : acquisition 3D utilisant le système 3D, l'axe en bleu permet de mesurer la position de la Tête à savoir Le « pitch » est l'angle entre le plan horizontal et le plan de Francfort. Le « Roll » est l'angle entre le plan horizontal et le plan bi-pupillaire. Enfin, le « Yaw » est l'angle entre le processus acromial et le plan sagittal médian.

IMPLICATION DE LA POSTURE DE LA NHP

Des études ont montré que la posture de la tête est corrélée à la morphologie crâniofaciale[3, 12], à la malocclusion[4], au schéma de croissance faciale[4], aux altérations respiratoires[13, 14] et ophtalmologiques[6, 15].

Cependant, ces études utilisent différentes méthodes pour mesurer les postures de la tête, ce qui rend difficile la comparaison des résultats entre les études. Il serait intéressant de trouver une technique simple, rapide, peu coûteuse et précise pour mesurer la posture de la tête sur les trois axes.

La plupart des méthodes disponibles et utilisées dans les précédentes études, pour mesurer la position naturelle de la tête (NHP) utilisent une mesure instantanée correspondant à une position à un moment précis. Cependant, on sait que la NHP est un équilibre de plusieurs positions ; elle n'est donc pas une position fixe et une variation inhérente est présente pour chaque personne. Les caractéristiques et les implications de cette variation ne sont pas entièrement comprises.

Il est possible que l'évaluation de la NHP sur une période plus longue puisse être d'un plus grand intérêt plutôt que de se limiter à la position de la tête à un moment précis. Ces informations à propos de la variation individuelle de la posture de la tête pourraient nous aider à mieux comprendre la nature de la NHP ainsi que son implication sur les autres systèmes.

OBJECTIF DES RECHERCHES

Notre objectif de la première étude est de mettre en place un système pour mesurer de la posture de la tête qui est fiable et simple d'utilisation. Pour ce faire, nous allons analyser la validité du système IMU in vitro, puis in vivo, pour mesurer la posture de la tête.

Nous voulions également estimer la variation de la posture de la tête dans un groupe de jeunes adultes en bonne santé pour mieux comprendre les caractéristiques de la position de tête.

Dans la deuxième étude, nous allons explorer les variations de la posture de la tête sur une période de 5 minutes. Pour ce faire, nous allons comparer l'analyse instantanée à une analyse de cinq minutes de la position de la tête. De plus, nous allons analyser l'évolution de la NHP dans une position auto-équilibrée pendant 5 minutes et la comparer à la position de la tête guidée par miroir pour les trois axes.

Enfin, nous allons évaluer la reproductibilité de la posture auto-équilibrée de la tête et de la position de la tête guidée par miroir pour les trois axes chez un groupe de jeunes adultes en bonne santé.

ARTICLE ORIGINAL 1



OPEN

Evaluation of head posture using an inertial measurement unit

Mustafa Al-Yassary^{1,3}✉, Kelly Billiaert^{1,3}, Gregory S. Antonarakis¹✉ & Stavros Kiliaridis^{1,2}

An inertial measurement unit (IMU) is an electronic device that measures and track the orientation of a body. We conducted this study in accordance with the STARD guidelines to evaluate the accuracy of IMU (index test) for measuring head posture compared to the current gold standard using a cervical range of motion (CROM) device. The reproducibility of the hunter and mirror-guided head posture was also evaluated. In vitro and in vivo tests were carried out to assess the validity of the IMU. To assess reproducibility, thirty healthy young adults were asked to look at four different locations in two different sessions while the head posture was recorded. Excellent correlation ($r = 0.99$; $p < 0.001$) was found between the IMU and CROM device with an absolute mean difference of $0.45^\circ \pm 0.58^\circ$ ($p = 0.85$) for the in vitro test and $0.88^\circ \pm 1.20^\circ$ ($p = 0.99$) for the in vivo test. For the reproducibility test, moderate to good correlation coefficients were found ($r = 0.55$ to 0.89 ; all $p < 0.05$) between the two sessions. The intraclass correlation coefficient ranged from moderate to excellent reliability (ICC from 0.74 to 0.96). These results suggest that the IMU sensors, when calibrated correctly, can be adequate to analyze head posture.

Natural head posture (NHP) is defined as “when a man is standing and when his visual axis is horizontal, his head is in a natural position”, first introduced by Broca¹. Studies have shown that the head posture has been correlated to craniofacial morphology^{2,3}, malocclusion⁴, facial growth pattern⁵, respiratory^{6,7}, and ophthalmology^{8,9} alterations. However, those studies use different methods to measure the head postures making it difficult to compare the results between studies. It would be interesting to find a simple, fast, inexpensive and precise technique to measure the posture of the head in the 3 axes.

Different methods exist to analyze head posture. The subjective method is an interpretation by the operator to assess the NHP¹⁰. For research purposes, where an objective assessment of head posture is sought, there are three types of methods to assess head posture:

Head-mounted equipment This is the standard method to assess head posture, which can be considered as the gold standard. The Cervical Range of Motion (CROM) device is a good example¹¹, but there can be faulty readings if the device has been placed incorrectly or if the patient has facial asymmetry. Moreover, when installing the device, the patient may adopt a different position due to the weight of the device. Head-mounted equipment can also be used with electronic gyroscopes.

Two-dimensional photographic analyses The use of photographs is another method to assess the NHP. It is simple, fast, inexpensive, and gives accurate results. Unlike the previous method, it does not require the patient to wear any bulky equipment. However, the axis of the photograph must be very precise, and it can be difficult to adopt with patients that are moving¹². The same principle can be used in radiology, but it exposes the patient to unnecessary radiation.

Three-dimensional (3D) analyses 3D optic scans of the head can also be used to measure the NHP. The patient does not have to wear any equipment and some systems have very high precision. The patient however has to stay still for a certain period without any movement, and this may introduce some bias. This method needs a complex setup, is expensive, and requires a good knowledge of the associated software¹³.

To perform reproducible analyses, there are three standards postures, the self-balanced, the self-guided, and the hunter position. The self-balanced head posture is adopted when a person looks at an imaginary point in front of them, from an infinite distance at eye level¹³. The self-guided head posture is when a person looks at themselves in a mirror². The hunter position is adopted when a person looks at a certain point⁸. The head posture can be evaluated along three axes: pitch, roll, and yaw.

In recent years, wearable sensors composed of an accelerometers and gyroscopes, commonly called inertial measurement units (IMUs), have become increasingly popular to measure human motion. IMU allows to

¹Division of Orthodontics, University Clinics of Dental Medicine, University of Geneva, Geneva, Switzerland. ²Dept of Orthodontics and Dentofacial Orthopaedics, School of Dental Medicine, University of Bern, Bern, Switzerland. ³These authors contributed equally: Mustafa Al-Yassary and Kelly Billiaert. ✉email: Mustafa.Al-Yassary@unige.ch; Gregory.Antonarakis@unige.ch

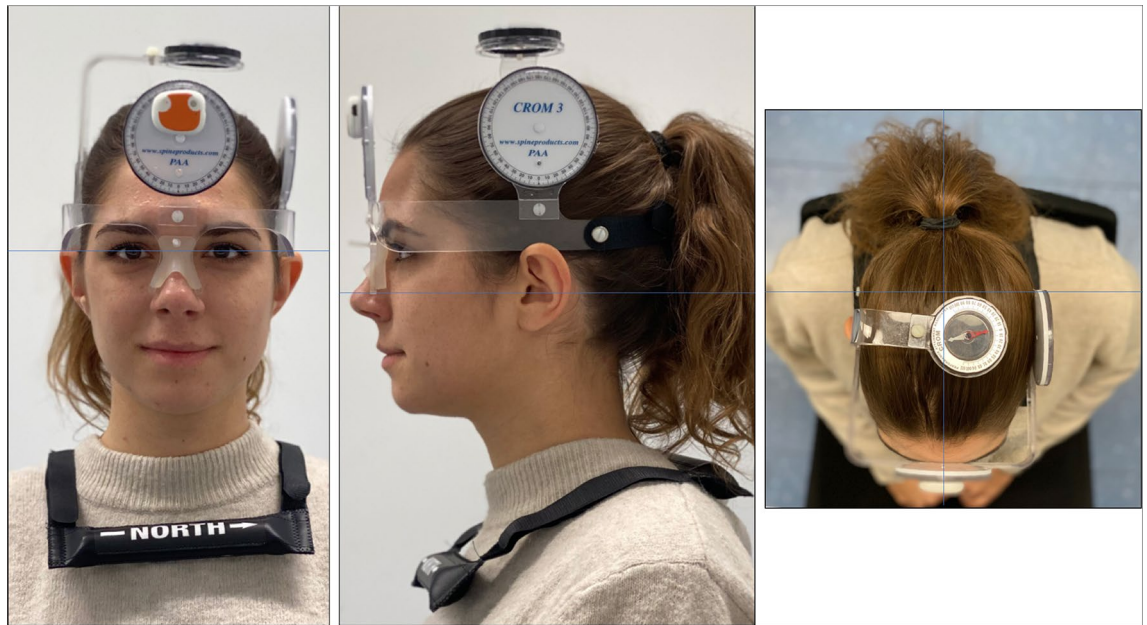


Figure 1. Photograph of the setup used for the in vivo assessment of validity of the inertial measurement unit. The cervical range of motion device was placed on the head with the first sensor fixed on the roll inclinometer and the second one placed on the ground acting as the reference.

measure joint angles using the orientation in space collected from two sensors^{14,15}. Magnetic angular rate and gyroscope sensors can also be part of the IMU, the additional reading of the earth's magnetic field allows to have more accurate orientation in 3D¹⁶. The benefits of this system are its ease of use, simplicity, and affordability. Our goal was first to analyze the validity of the IMU system in vitro and then in vivo for measuring the head posture. We also wanted to estimate the variation of head posture in a group of healthy young adults.

Methods

Ethics statement. The present study was approved by the Swiss Association of Research Ethics Committees. The experimental procedures were conducted in conformity with the Declaration of Helsinki. Informed consent for participation in the study and publication in an open access format was obtained from the participants, with regard to their photographs and personal information. The procedures of the study were fully explained to the participants and they provided their informed written consent before testing.

Material. For the present study, the MetaMotionR (MbleintlabInc., San Francisco, CA, USA) was used as the IMU. This system is composed of two sensors. This system uses cloud storage to allow health professionals to monitor the patients. Calibration is required prior to use, and this is carried out by positioning the sensor in a neutral position (corresponding to 0° on each axis).

We followed the recommendation of the STARD guidelines¹⁷ for reporting the diagnostic accuracy of the index test (IMU) compared against the reference standard (CROM device). Investigations of validity were performed both using in vitro and in vivo models. The CROM device is considered the best available method to date to assess head posture (gold standard). This device is placed on the head like glasses, and includes two inclinometers, one located on the forehead and the other on the temporal bone. This allows for the assessment of pitch and roll. A compass is located at the top of the head and along with the magnets on the shoulders this allows for the assessment of the yaw (Fig. 1).

In vitro assessment of validity. The validity of the method was first investigated using an in vitro model. A box-shape model was used as the representation of the head, which could be moved and blocked on the three axes separately. Initial calibration of the IMU was carried out when the readings from the CROM device on the three axes were 0°. Two IMU sensors were used, one placed on the front inclinometer and the second placed on a table behind it (Fig. 2). The model was subsequently moved, one axis at a time, guided by the CROM device. The degrees were increased by 10° sequentially until 90° were reached. The same methodology was used for the negative degrees (from 0° to -90°). At each 10° interval, measurements were recorded using the IMU. The in vitro assessment was carried out on two separate occasions, with a one-week interval between sessions. In total 60 recordings (20 for each of the three axes) were taken with the CROM device and the IMU at each session.

In vivo assessment of validity. The validity of the method was investigated with five healthy collaborators at the University clinics of dental medicine in Geneva, Switzerland. The human subjects wore the CROM device on their head and the IMU sensor was placed on the front inclinometer. The second sensor was placed on the floor to act as the reference, using the same concept as in the in vitro model. The IMU was calibrated when the

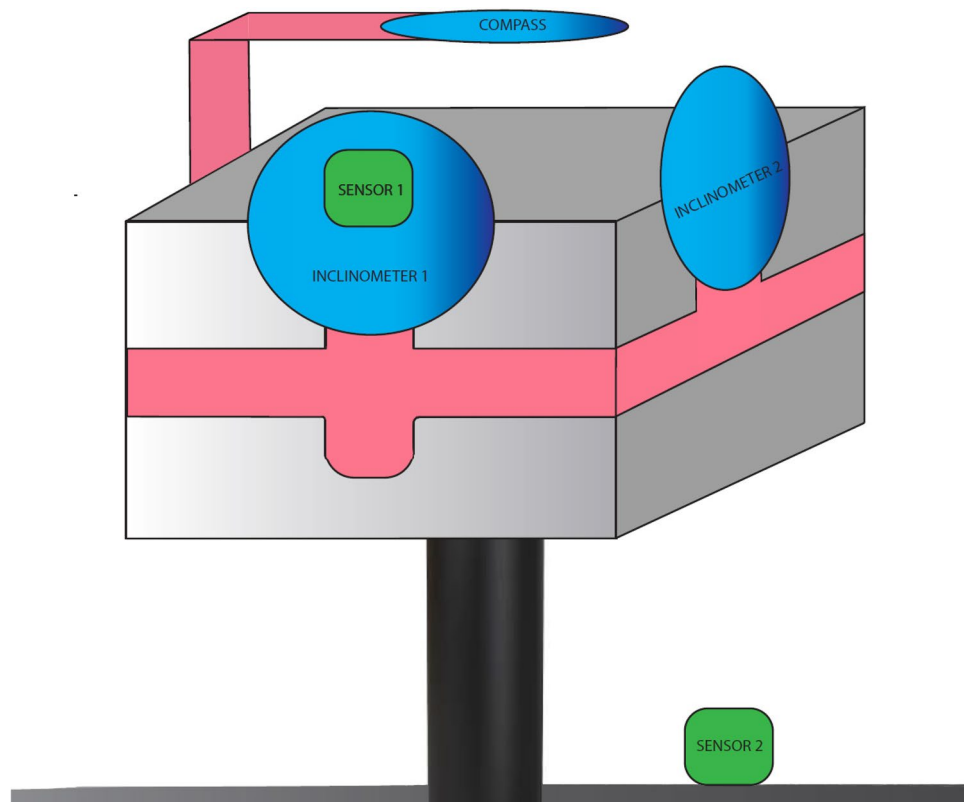


Figure 2. Representative image for the in vitro assessment of the inertial measurement unit. A box-shaped model (grey) was used with the cervical range of motion device (pink and blue) placed on top. The first sensor (green) was placed on the inclinometer 1, and the second on the table. It was possible to move and block this model on the 3 axes separately.

readings from the CROM device on the three axes were 0°. The subjects were subsequently asked to move their head on each axis separately. The analysis was carried out with a sequential increase in each axis by 10° up to the maximum. At each interval, the reading from the IMU was recorded (Fig. 1). In total, 217 recordings were transcribed (ranging from 48 to 39 recordings per person).

Variation of head posture and reproducibility assessment. A group of 30 healthy young adults, all collaborators of the University clinics of dental medicine in Geneva, Switzerland, were used to assess the variation of head posture and the reproducibility of the method. They were asked to look at four different locations, while standing (orthoposition¹⁸) in a well-illuminated room. This included a mirror located three meters in front of them where they were asked to look at themselves, as well as three different points which were indicated with a number and a point (0.5 cm of diameter) (Fig. 3). Head posture was recorded with the IMU as the participant looked at each of these four locations. Calibration of the IMU was carried out with the participants looking at their eyes in the mirror (mirror guided head posture), with one sensor on the forehead and the other on the floor used as the reference. Following this calibration, the participants were asked to look at each of the three points in order, looking at the mirror in-between each of the three positions. This was done twice for each position during the same session. The same procedure was carried out on two separate occasions, with a one-week interval. In total 20 recordings were carried out for each participant (10 in each session, namely two for each position and four for the mirror).

Statistical analyses. All statistical analyses were performed using SPSS (version 24.0, SPSS Inc., Chicago, IL, USA). To compare the results of the IMU system with the CROM device, paired t-tests were used as well as Pearson correlation coefficients for each position. Linear regression analysis was performed for the in vitro and in vivo test with the results from the CROM device as the independent variable, and those from the IMU as dependent variable. Intraclass correlation¹⁹ estimates and their 95% confident intervals (CI) were calculated²⁰. ICCs performed for the in vivo and in vitro tests were based on a single rating, absolute-agreement, 2-way random-effects model, while for the variation in head posture and reproducibility ICCs were performed based on a mean-rating (k = 4), absolute-agreement, 2-way mixed-effects model. ICC values less than 0.5 are indicative of poor reliability, values between 0.5 and 0.75 indicate moderate reliability, values between 0.75 and 0.9 indicate good reliability, and values greater than 0.9 indicate excellent reliability²¹. The standard error measurement (SEM) of the difference between session 1 and 2 was calculated for each ICC (using the formula $SEM = StandardDeviation \times \sqrt{1 - ICC}$)²⁰.

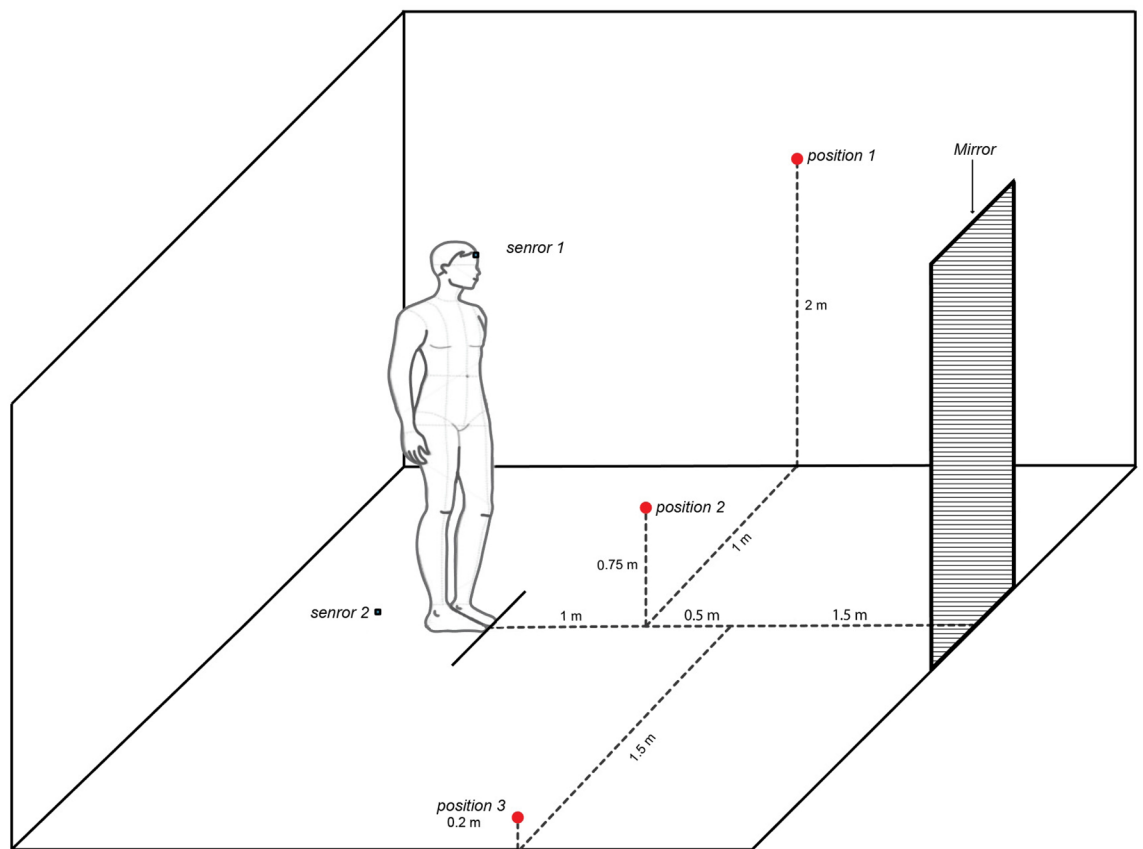


Figure 3. Representative image for the assessment of head posture and the reproducibility of the inertial measurement unit system. The participants were positioned in front of a mirror located 3 m in front of them. The first point was located 90 cm in front, 1 m to the left, and 2 m above their feet (position 1). The second point was 90 cm in front and 75 cm in height (position 2). The third point was located 1.50 m in front, 1.50 m to the right, and 20 cm in height (position 3).

For the in vivo test, a Bland–Altman plot was made to visualize the spread of the differences between the CROM device and the IMU for each recording compared to their mean. Bland–Altman plots were also used to visualize the spread of the differences between the two sessions for each individual axis.

Results

In vitro assessment of validity. The mean of the absolute difference between the CROM device and the IMU in vitro was $0.45^\circ \pm 0.58^\circ$ ($p=0.85$). A strong correlation was observed between the two systems ($r=0.99$, $p<0.001$). For every axis and between the two sessions, no systematic differences between the two systems were reported ($p>0.05$). The ICC between the CROM device and IMU was excellent ($ICC=1.000$) with a 95% CI between 0.999 and 1.000 and a SEM of 0° .

In vivo assessment of validity. The mean of the absolute difference between the CROM device and the IMU in vivo was $0.88^\circ \pm 1.20^\circ$ ($p=0.99$). A strong correlation was observed between them ($r=0.99$, $p<0.001$). For each axis no systematic differences between the two systems were reported ($p>0.05$). Unlike for the in vitro study the yaw was slightly less precise than the pitch and the roll (Table 1). Even with extreme head postures, an increase in the variation between the CROM device and the IMU was not observed. The ICC between the CROM device and the IMU was excellent ($ICC=1.000$) with a 95% IC between 0.999 and 1.000 and a SEM = 0° .

The Bland–Altman plot shows the mean differences at $+0.2^\circ$ and the 95% CI between -2.15 and $+2.54$. No systematic bias is observed, with the data points being distributed equally below and above the mean line. Most of the points situated outside the 95% CI are Yaw measurements (Fig. 4).

Variation of head posture and reproducibility. The most stable axis (with the smallest SD) was found to be the Roll, followed by Yaw and finally Pitch (Table 2). For every axis and between the two sessions, no systematic error was found ($p>0.05$). The ICC between all eight recordings for the mirror position had moderate to good reliability (ICC between 0.74 and 0.96) depending on the axis (Table 2).

The Bland–Altman plot shows the distribution of the difference between the two sessions. The mean of the first four recordings for the mirror position were calculated and compared to the mean of the last four recordings. The X axis represent the mean between the two sessions and the Y axis the differences between them (Fig. 5).

	Total	Pitch	Roll	Yaw
In vitro				
Mean $\pm$ SD	0.45 $\pm$ 0.58	0.47 $\pm$ 0.62	0.38 $\pm$ 0.50	0.50 $\pm$ 0.49
r (p value)	0.99 (0 < 0.001)	0.99 (< 0.001)	0.99 (< 0.001)	0.99 (< 0.001)
In vivo				
Mean $\pm$ SD	0.88 $\pm$ 1.20	0.61 $\pm$ 0.80	0.80 $\pm$ 1.04	1.28 $\pm$ 1.61
r (p value)	0.99 (< 0.001)	0.99 (< 0.001)	0.99 (< 0.001)	0.99 (< 0.001)

Table 1. Summary table comparing the inertial measurement unit to the cervical range of motion device. The mean and standard deviation (SD) of the differences represents the average of the absolute difference between the cervical range of motion device (gold standard) and the inertial measurement unit. The Pearson correlation (r) with the respective p value is also shown.

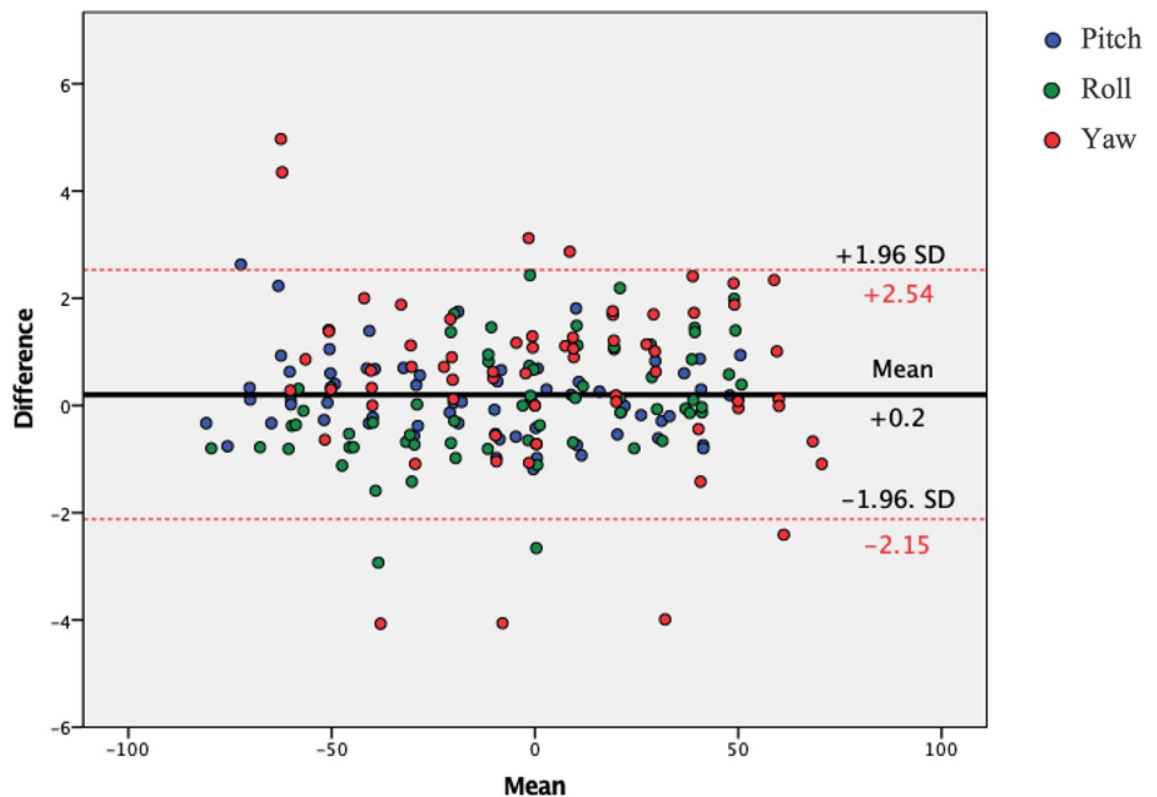


Figure 4. Bland–Altman plot between the cervical range of motion device and inertial measurement unit in vivo. The X axis represents the mean between the cervical range of motion device and the inertial measurement unit and the Y axis the mean differences between them. Red lines indicate the 95% confidence intervals, and the black line indicates the mean of the differences. Blue points represent the Pitch, green points the Roll and red points the Yaw.

The mirror position shows a higher stability than the hunter position with 95% CI < 3° for the mirror and 95% CI > 5.5° for the hunter positions. Overall, the distribution is equal above and below the mean.

For the hunter position, four recordings were measured for each position (two during each session). A greater SD was observed compared to the mirror position but with better reliability (from moderate to excellent) (Table 2). Overall, there was no systematic difference between the first and second session ($p > 0.05$).

To better understand the stability and reproducibility specific to each participant the measurements from the two sessions were compared. For the three hunter positions, the difference between the mean of the first two recordings (done the same day) and the last two recordings (done after one week) was calculated. Significant intra-individual variation was observed. The roll seems to be the most precise and stable axis, followed by pitch, and finally yaw with the largest amount of variation (Table 3). Paired T-test showed no systematic difference between the first and second sessions ($p > 0.05$).

	Mean $\pm$ SD	ICC	95% CI	SEM
Mirror				
Pitch	-0.31 ± 2.34	0.74	0.57–0.86	1.20
Roll	0.35 ± 1.22	0.82	0.70–0.90	0.52
Yaw	0.41 ± 1.83	0.84	0.74–0.91	0.73
Position 1				
Pitch	-7.00 ± 3.98	0.89	0.81–0.94	1.32
Roll	-2.65 ± 4.30	0.92	0.86–0.96	1.24
Yaw	32.88 ± 5.35	0.73	0.53–0.86	2.77
Position 2				
Pitch	26.48 ± 7.54	0.95	0.91–0.97	1.75
Roll	1.62 ± 2.36	0.85	0.73–0.92	0.92
Yaw	-3.57 ± 2.45	0.83	0.71–0.91	1.00
Position 3				
Pitch	15.17 ± 5.67	0.96	0.92–0.98	1.19
Roll	-7.71 ± 4.44	0.93	0.89–0.97	1.15
Yaw	-36.41 ± 5.72	0.94	0.90–0.97	1.39

Table 2. Characteristics of the natural head position achieved in front of a mirror and each of three hunter positions. Means and standard deviations (SD) are shown for each position and each axis within each position, for the 30 participants. Intraclass correlation¹⁹, 95% confidence intervals (CI), and standard error measurements (SEM) are also shown.

Discussion

Assessment of validity. The present study has shown that there is an excellent correlation in vitro as well as in vivo between the CROM device and the IMU. Our results on head posture are similar to the ones found previously by Beange et al.²² who compare the MetaMotionR to a 7-camera motion capture (Vicon MX40) on the spine with an excellent degree of similarity. The yaw is slightly less precise than the two other axes. It is mainly due to the fact that the pitch and the roll are given by two inclinometers but the yaw is given by compass on the CROM device. The compass has a larger scale (every 2° for the compass and every 1° on the inclinometers).

The difference between the two systems is slightly bigger in the in vivo test. Two important potential reasons are that the patient cannot remain completely stable, and that the data readings given by the CROM device are more complicated to report. These reasons are operator and patient dependent and are not directly related to the sensors themselves. After using both systems, the authors feel that the IMUs are easier to use because they allow to measure the head posture on the 3 axes simultaneously and do not rely on the human readings of the data and thus eliminate bias in this regard. Furthermore, the sensors are very light weight and not bulky, unlike the CROM device, and this allows measurements of the natural position of the patient to not be influenced by the equipment worn. However, unlike the CROM that can be used directly on the patient's head, the IMU system needs to be calibrated before the measurement in order to give accurate results.

Assessment of reproducibility. When looking at a point (hunter's position), three main parameters may influence the posture of the head, namely the posture of the body²³, the gaze of the eyes^{24,25}, and location in the field of vision²⁶. When asking the participant to maintain a straight body posture, without moving their eyes and keeping the point in the center of the field of view, it is likely that small head and eye movements can still take place which can explain variations between the two sessions. This variation between sessions is however interesting to study. It was observed that participants with a poor correlation in the first session also had a poor correlation in the second session and vice versa. This can lead to hypothesize that the stability of the posture of the head is very specific to each individual. In addition, it was noticed that in a group of healthy patients, the variation is greater with the rotation and the pitch, but the roll is very stable. Regarding the mirror-guided posture, our results are very similar to previous studies regarding the pitch. Solow and Tallgren² found that the NHP which refers to the mirror-guided head posture is very reproducible with a range of 1.4°. That is very close to our findings (with a range of 2.3°). We also noticed that there are variations detected for the roll and yaw axes.

If we take into consideration the different factor that can influence head posture during the hunter position, we can affirm that this IMU system has a good reproducibility compatible to carry out advanced research in the field.

Limitations. These results must be interpreted with caution because the population of our study concerns only healthy young adults, with an analysis of NHP in a standing position. It would be interesting to evaluate the posture of the head in a larger population and with different position (such as the sitting position for example).

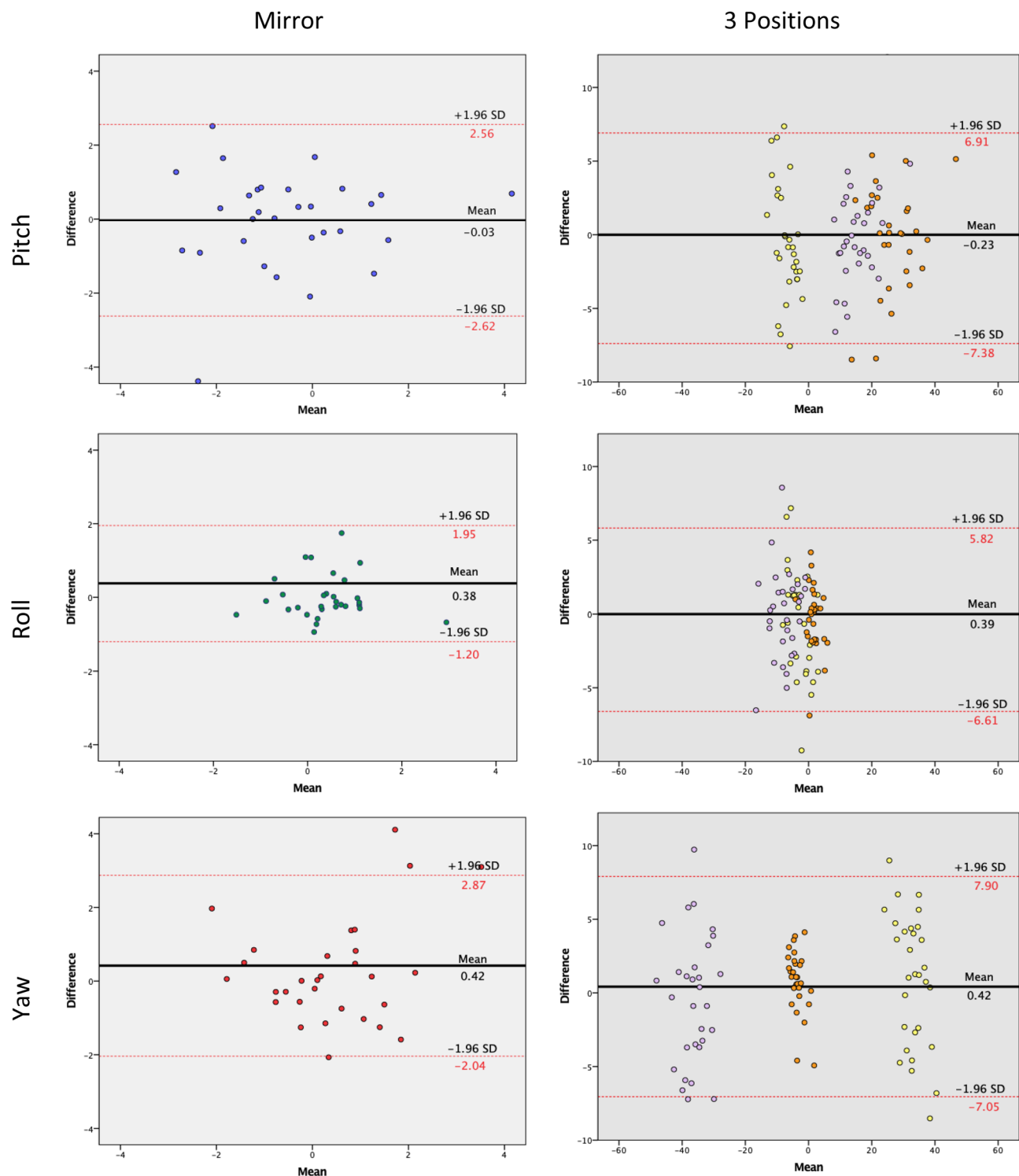


Figure 5. Bland–Altman plot between the two session: for mirror and the three positions. Bland–Altman plots show the mean differences in the measurement of the natural head position using the mirror (left column) and the 3 hunter positions (right column) from the 30 participants. Red lines indicate the 95% confidence intervals while black line indicate mean of the differences. For the 3 positions, yellow points indicate the first position, orange points the second position and purple points the third position.

Conclusion

IMU sensors have an excellent correlation with the physical head-mounted equipment (CROM device) both for in vitro and in vivo tests. When calibrated correctly, IMU sensors can be a reliable method to analyze head

Position		Mean $\pm$ SD	r (p value)
1	Pitch	2.41 $\pm$ 3.31	0.64 (<0.001)
	Roll	2.60 $\pm$ 3.40	0.70 (<0.001)
	Yaw	3.90 $\pm$ 4.49	0.57 (<0.001)
2	Pitch	2.97 $\pm$ 4.19	0.84 (<0.001)
	Roll	1.56 $\pm$ 2.14	0.55 (0.002)
	Yaw	1.78 $\pm$ 2.11	0.61 (<0.001)
3	Pitch	2.27 $\pm$ 2.83	0.89 (<0.001)
	Roll	2.33 $\pm$ 3.05	0.75 (<0.001)
	Yaw	3.53 $\pm$ 4.35	0.66 (<0.001)

Table 3. Difference between the two sessions Differences, presented as absolute mean and standard deviation (SD), between the mean of the 1st and 2nd recording (done in the first session) and the mean of the 3rd and 4th recording (done in the second session) for each axis are shown. Pearson correlation coefficients (r) and p value are also shown.

posture. Variations in NHP are approximatively 2° for the pitch and slightly less for the roll and the yaw axes. Stability of head posture is specific to each person and varies depending on the axis.

Data availability

All data included in this study are available upon reasonable request from the corresponding author.

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Figure 2 and 3 have been drawn by M.A. and K.B. using adobe illustrator (version 24.3, cc 2020) (<https://www.adobe.com>). Figure 1 is a photograph taken by M.A. of a participant who agrees to be published.

Author contributions

M.A., K.B. contributed to the data collection and data analysis, G.S.A., S.K. contributed to the study design and drafting of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.A.-Y. or G.S.A.

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ARTICLE ORIGINAL 2

Evaluation of natural head position over five minutes: A comparison between an instantaneous and a five-minute analysis with an inertial measurement unit

Mustafa Al-Yassary  | Kelly Billiaert  | Gregory S. Antonarakis | Stavros Kiliaridis

Division of Orthodontics, University
Clinics of Dental Medicine, University of
Geneva, Geneva, Switzerland

Correspondence

Mustafa Al-Yassary and Gregory S.
Antonarakis, Division of Orthodontics,
University Clinics of Dental Medicine,
University of Geneva, Geneva,
Switzerland.
Emails: Mustafa.Al-Yassary@unige.ch
(MA); gregory.antonarakis@unige.ch (GA)

Abstract

Background: Head posture is a balance of several positions and therefore shows inherent variation. Most methods available to quantify this are however instantaneous, not providing information about its variation over time. A dynamic recording of head posture would thus be beneficial.

Objectives: The purpose of this study was to evaluate the variation in natural head position (NHP) over 5 min using an inertial measurement unit (IMU).

Methods: Fifteen healthy young volunteers were asked to sit on a chair and keep their head in the self-balanced position for 5 min. A mirror was then revealed in front of them, and they were asked to look at their eyes for 20 s. This procedure was undertaken on two separate occasions with a one-week interval. This was compared to an instantaneous measurement of head position at a specific time point corresponding to the 15th second of the recording.

Results: During the 5 min of recording, the participants tended to elevate their head progressively by a mean of 1.5°, which is then corrected by looking at oneself in the mirror. Most participants tended to rotate their head to the left and continued that progressive rotation despite looking in the mirror. The roll axis had no systematic changes observed between the self-balanced position and the mirror-guided position and was the most reproducible axis. Moderate to good correlations were found comparing both sessions for each axis.

Conclusion: The comparison between the five-minute analysis and the instantaneous measurement showed a systematic difference on the pitch axis but no differences for the yaw and roll. These results suggest that the variation in the NHP during a period of 5 min is generally specific to each participant with a head elevation and rotation to the left in most cases.

KEYWORDS

Clinicians, head, position, sitting, wearable electronic devices

Mustafa Al-Yassary and Kelly Billiaert contributed equally to this work.

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1 | INTRODUCTION

Most methods available to measure the natural head position (NHP)¹ use an instantaneous measurement corresponding to one position at a specific point in time. However, the NHP is known to be a balance of several positions²; therefore, it is not a fixed position, and inherent variation is present for each person. The characteristics and implications of this variation are not fully understood. The head position is related to different alterations in fields such as orthodontics,³ orthopaedics,⁴ respiratory physiology,^{5,6} and ophthalmology.⁷

To measure the NHP, there are two main standard positions, namely, the self-balanced position and the self-guided position. The self-balanced position is adopted when a person looks at an imaginary point in front of him/her, from an infinite distance at the eye level.⁸ The self-guided position is when a person looks at themselves in a mirror.⁸ Different studies indicate that the mirror-guided head position is the most stable and reproducible^{9,10}; however, it is rarely adopted in everyday life. It is thus preferable, in order to more accurately simulate the head position in a natural environment, to analyse the self-balanced head position that is more frequently adopted. In growing individuals, there are hypotheses that the head position may influence the craniofacial growth pattern.^{8,11} In such cases, it is possible that the evaluation of the NHP during a longer period could be of greater interest rather than solely looking at the head position at a specific point in time.

Evaluation of the NHP is performed on three axes namely pitch, roll, and yaw. The pitch is the angle between the horizontal plane and the Frankfort plane. The roll is the angle between the horizontal plane and the bipupillary plane. Finally, the yaw is the angle between the acromial process and the median sagittal plane. (Figure 1).

The aims of this study were to

1. Compare the instantaneous analysis to a five-minute analysis of the head position.
2. Analyse the evolution of the NHP in a self-balanced position for 5 min and compare it to the mirror-guided head position for the three axes.
3. Evaluate the reproducibility of the self-balanced head posture and the mirror-guided head position for the three axes in a group of healthy young adults.

2 | METHODS

2.1 | Ethics statement

The present study was approved by the Swiss Association of Research Ethics Committee. The experimental procedures were conducted in conformity with the Declaration of Helsinki. Informed consent for participation in the study and publication in an open-access format was obtained from the participants, with regard to their recordings and personal information. The procedures of the

study were fully explained to the participants, and they provided their informed written consent prior to being enrolled in the study.

2.2 | Subjects and methods

Fifteen healthy young adult volunteers (eight women and seven men) at the University Clinics of Dental Medicine in Geneva, Switzerland, aged from 20–30 years (mean age of 28.4 years) were recruited for this study.

Wearable sensors or inertial measurement units (IMU)s were used to measure head position. These are electronic devices allowing movements to be tracked in three dimensions (3D).¹² This system was chosen because it is easy to use, precise,¹³ and allows tracking of head position in a given period. This system is composed of a gyroscope, an accelerometer, and a magnetic angular rate and gravity. The hardware consists generally of two parts, a detector fixed on the element of interest and a receiver. The information is stored to allow one to access the recordings of the patients and to follow them.¹⁴ An IMU allows the head position to be recorded dynamically for a prolonged time period, thus it is possible to analyse the full variation of the NHP during a 5-minute recording.

The participants were asked to sit on a chair in the 90–90–90 position (corresponding to a 90° angle on the hips, the knees, and the feet)¹⁵ facing a white wall which was one meter in front of them. They were asked to look in front of them and to keep a straight head posture with their own feeling of natural head balance (self-balanced position). This position was recorded for 5 min while listening to a short story played behind them with stereo-speakers carefully centred on the volunteers. Following this, a mirror was placed on the wall in front of the participants, and they were asked to look at their eyes for 20 s (self-guided position). This procedure was repeated twice with a one-week interval, and during the second recording, a different short story was played in order to allow for a similar experimental setup without the participants having to listen to the same story twice.

The IMU system used was the MetaMotionR (Mbleintlab Inc.). This system is composed of two sensors. Before measuring the head position, this system needs to be calibrated to assess the neutral position (corresponding to 0° on each axis). This was done by placing the first sensor on the ground in front of the participant and the second one behind them. Both have to be parallel, on the same plane, and perpendicular to the ground axis. After calibration, the first sensor was placed on the forehead of the participant.

In order to minimize the error due to the positioning of the sensor, one front face photograph was taken to make sure that the sensor was parallel to the bipupillary line. A second profile photograph was taken to eliminate the slope of the forehead from the IMU recordings. The postprocessing calibration was performed with Pixelstick (version 2.16.2, Plum Amazing software LLC). This calibration method was the same as that used by Billiaert et al.¹⁶ that compared the IMU system to lateral photographs and showed excellent accuracy.

FIGURE 1 Representation of head movements: showing the representation of the head movements on the three axes, namely, the pitch axis, roll axis, and yaw axis

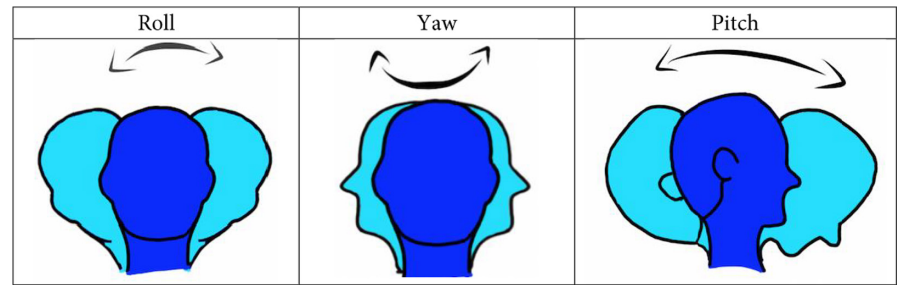


TABLE 1 Summary of the first session of the study population head position measured with an inertial measurement unit

N = 15	Pitch	Roll	Yaw
Mean $\pm$ SD	$-16.8^{\circ} \pm 4.6$	$-1.5^{\circ} \pm 3.2$	$1.5^{\circ} \pm 3.0$
95% CI	$-19.1 - -14.5$	$-3.1 - 0.1$	$0 - 3.0$

Note: The recordings correspond to an instantaneous analysis of the head position in a sitting and self-balanced position. Shown are the number participant (n), the mean, standard deviation (SD), and 95% confidence interval (95% CI) of the recordings for each axis.

2.3 | Statistical analysis

All statistical analyses were performed using SPSS (version 24.0, SPSS Inc.). The data were separated into four different segments. The initial segment (I) represents the mean of the first 20 s, the finishing segment (F) the mean of the last 20 s before the mirror-guided position. The self-guided position (S) is the mean of the full segment from initial to finish. Finally, the mirror-guided position segment (M) represents the mean of the 20 s when the mirror was revealed in front of the participants (Figure 2). The means of 20 s were taken for the I, F, and M segments in order to have a more reliable representation of the head position. In total, for each participant, 24 means were calculated (two sessions, four segments for each session, and three axes for each segment). The extreme movements (outliers) that were not related to the NHP (for example if a participant sneezes, coughs, or moves their arms or feet) were eliminated.

For the instantaneous measurement, recorded data were taken at the 15th second with the IMU system. As there are no clear instructions in previous studies regarding the time at which the photograph for the head position measurement was taken, the 15th second was chosen because it represents approximatively the time required to take a photograph.

The differences between the instantaneous measurement and the five-minute recordings were analysed. The instantaneous measurement was taken and compared to the mean of the recordings for the self-balanced head position (S). The difference between I-F, F-M, and I-M was analysed to better understand the evolution and tendency of the head position. The self-guided head position (S) was also compared to the mirror-guided head position (M), and the two sessions were compared to assess whether the variation within the five-minute recording period was random or if it followed a specific and reproducible pattern.

For each set of measurements, the difference between the means was calculated, with *p* values inferior to .05 considered statistically significant. Pearson's correlations and intraclass correlation (ICC)*¹⁷ with 95% confidence intervals (95% CI) (based on a mean-rating (*k* = 2), absolute-agreement, 2-way mixed-effects model) were also calculated. ICC values less than 0.5 are indicative of poor reliability, values between 0.5 and 0.75 indicate moderate reliability, values between 0.75 and 0.9 indicate good reliability, and values greater than 0.90 indicate excellent reliability.¹⁸ The standard error measurement (SEM) was calculated for each ICC (using the formula $SEM = Standard\ Deviation \times \sqrt{1 - ICC}$).

3 | RESULTS

The data show a greater interindividual variation in the pitch axis with a negative value corresponding to an elevated Frankfort plane than the horizontal plane. The roll axis had the least amount of interindividual variation and a slight deviation to the right compared to the median plane. The yaw axis had comparable interindividual variation to the roll axis and had a slight deviation to the left compared to the plane that passes by the acromial process (shoulders) (Table 1).

3.1 | Comparison of the one-shot procedure to the five-minute analysis

When comparing the instantaneous analysis to the mean of the five-minute analysis, a systematic difference was found for the pitch axis (*p* = .04), whereby the mean pitch from the five-minute procedure was elevated by 0.9 degrees compared to the instantaneous procedure. For the roll and yaw axes, no systematic differences (*p* > .05) were found. However, a good correlation and good to excellent reliability was found for all the three axes (Table 2).

3.2 | Analysis of the evolution of the NHP in a self-balanced position for five minutes compared with the mirror-guided head position for the three axes

During the self-balanced position, the participants tend to elevate their head by $1.5^{\circ} \pm 3.8^{\circ}$ (*p* = .03), which was then corrected by the mirror-guided position ($-1.3^{\circ} \pm 4.2^{\circ}$; *p* = .013). The roll axis has the

	Pitch	Roll	Yaw
Mean $\pm$ SD (p-Value)	0.9° $\pm$ 2.2 (0.04)	0.1° $\pm$ 1.2 (0.85)	-0.6° $\pm$ 2.7 (0.2)
r (p-Value)	0.93 (0.00)	0.92 (0.00)	0.79 (0.00)
ICC (95% CI)	0.95 (0.89–0.98)	0.96 (0.91–0.98)	0.86 (0.71–0.93)
SEM	0.48	0.25	1.00

Note: Shown are the mean and standard deviation (SD) of the difference ($T^1 - T^2$) between the instantaneous (corresponding to T^1) and the five-minute recording (corresponding to T^2), Pearson's correlation (r), the intraclass correlation coefficient (ICC)¹⁷ with 95% confidence interval (95% CI), and their standard error measurement (SEM) for each axis.

TABLE 2 Comparison between an instantaneous and a five-minute recording of head position using an inertial measurement unit

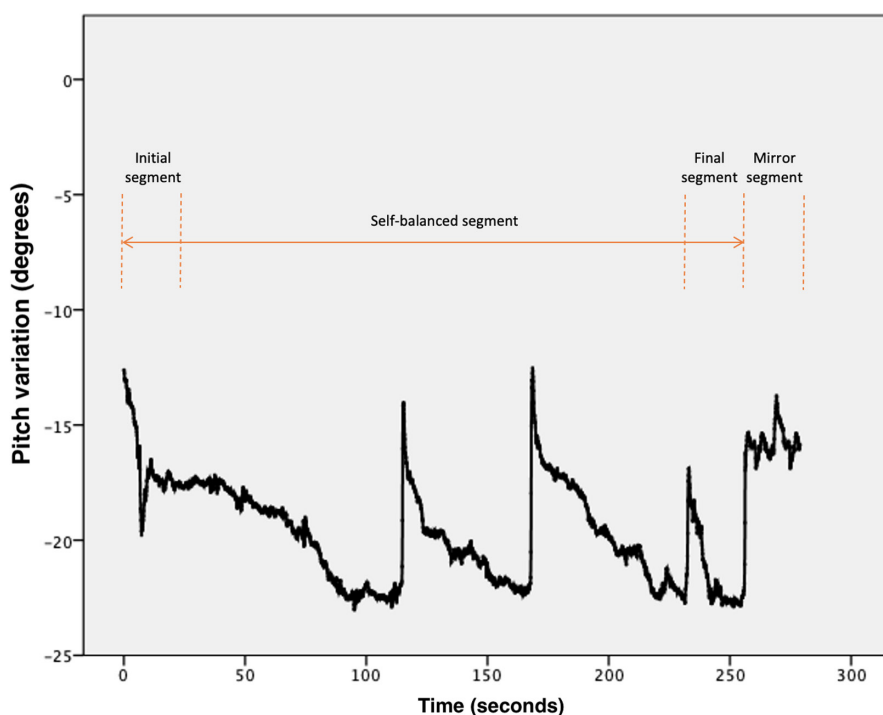


FIGURE 2 Representation of the variation of the natural head position on the pitch axis for five minutes showing the variation of head position for the pitch axis. The orange arrow represents the self-balanced segment of the recording with the initial segment at the beginning and the final segment at the end. The mirror segment (mirror-guided position) is represented following the self-balanced segment

least amount variation with the standard deviation ranging from 0.9° to 2.3°. On the yaw axis, the participants tend to progressively increase the deviation, generally to the left by $2.3^\circ \pm 4.7^\circ$ ($p = .13$), and the mirror-guided position does not seem to correct this.

Overall, great correlation was found between the different segments for the pitch and roll axes ranging from 0.72 to 0.87. However, for the yaw axis, the correlations were lower, ranging from 0.49 to 0.93 due to higher deviations. For the pitch and roll axes, good-to-excellent reliability was found between the segments (ranging from 0.83 to 0.93), and for the yaw axis moderate to excellent reliability was found (ranging from 0.55 to 0.96).

3.3 | Evaluation of the reproducibility of the self-balanced and the mirror-guided head position

No systematic differences were found between the two sessions for each segment ($p > .05$). When comparing the same segments for the two different sessions (Figure 3), moderate to good correlations were found depending on the axis. The pitch axis had the strongest correlations followed by the roll and the least reproducible axis was

the yaw. Good to excellent reliability was found for the self-balanced head position (ranging from 0.78 to 0.96) and poor-to-excellent reliability for the mirror-guided head position (ranging from 0.49 to 0.92) depending on the axis (Table 3).

4 | DISCUSSION

The present study showed an excellent correlation between the instantaneous procedure and the mean of the five-minute recordings for the pitch and roll axes and good correlation for the yaw axis. The five-minute analysis showed a systematic elevation of the head compared to the instantaneous procedure, but no systematic changes for the roll and yaw axes were observed.

The data recorded at the 15th second with the IMU system were used as a representative instantaneous procedure. This time was chosen because it represents approximately the time required to take a photograph. Moreover, unlike photography, the IMU system allows one to obtain data for all three axes. The study of Billiaert et al.¹⁶ as well as Al-Yassary et al.¹⁹ demonstrated excellent accuracy of the IMU system for measuring head position.

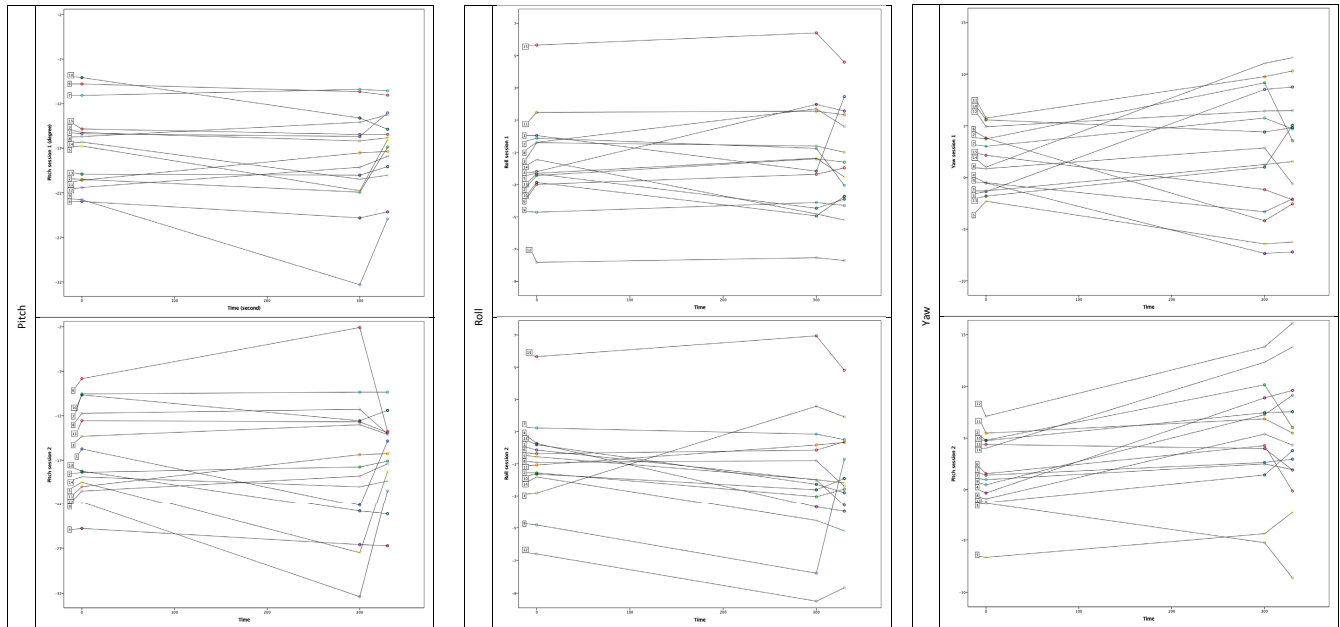


FIGURE 3 Spaghetti plots showing the evolution of the head position over a five-minute recording period for the three axes: The X-axis represents the time of recording in seconds. The Y-axis represents the intensity of the changes of the head position. For each participant, three means are represented by points on this plot (the initial segment corresponding to the mean of the first 20 seconds of the recording, the final segment corresponding to the mean of the last 20 seconds before the mirror-guided position, and the mirror segment corresponding to the mean of the mirror-guided segment)

TABLE 3 Comparison between the two sessions of the self-balanced head position and the mirror-guided head position

Segments		Pitch	Roll	Yaw
Self- balanced	Mean $\pm$ SD (<i>p</i> -Value)	$-0.6^{\circ} \pm 2.2$ (0.35)	$-0.3^{\circ} \pm 1.6$ (0.53)	$1.6^{\circ} \pm 3.5$ (0.11)
	<i>r</i> (<i>p</i> -Value)	0.90 (0.00)	0.72 (0.00)	0.69 (0.01)
	ICC (CI 95%)	0.96 (0.89–0.99)	0.94 (0.82–0.98)	0.78 (0.38–0.93)
	SEM	0.43	1.39	1.65
Mirror	Mean $\pm$ SD (<i>p</i> -Value)	$0.0^{\circ} \pm 2.4$ (0.96)	$0.1^{\circ} \pm 2.0$ (0.89)	$-2.4^{\circ} \pm 6.9$ (0.21)
	<i>r</i> (<i>p</i> -Value)	0.85 (0.00)	0.58 (0.24)	0.34 (0.22)
	ICC (CI 95%)	0.92 (0.76–0.97)	0.75 (0.21–0.92)	0.49 (0.41–0.82)
	SEM	0.68	1.22	4.93

Note: The self-balanced segment represents the mean of the five-minute recording of the head position in the self-balanced head posture, and the mirror segment represents the mean of a 20-second recording for the mirror-guided head posture. For each segment and axis, the first to the second session was compared. Shown are the mean and standard deviation (SD) of the difference between the two sessions, Pearson's correlation coefficient (*r*), the intraclass correlation coefficient (ICC)¹⁷ with 95% confidence interval (95% CI), and their standard error measurement (SEM) for each axis.

When the different segments were compared, it was found that the participants tended to elevate their heads and turn progressively to the left. However, a general trend for the roll axis was not found. When the mirror was displayed, it was observed that the participants corrected their elevation on the pitch axis, without any effects on the other two axes (roll and yaw). The participants that tended to rotate their head kept their rotation and continued in the same direction.

The variation of head position seems to be influenced by different factors. The rotation of the head may be related to the dominant eye, which is often the right eye. This observation is described by Pradham et al. who found a strong relationship between eye

dominance and direction of head rotations.²⁰ It was also observed that a progressive elevation in the self-balanced head position occurs over time to a maximum of 1.5° after 5 min, likely caused by muscle fatigue. This elevation of the head in the self-balanced head position compared to the mirror-guided head position has also been observed by Jakobsone et al.⁹

When comparing the two sessions, a stronger correlation was found for the self-balanced head position than the mirror-guided head position on each axis. The correlations of the mirror-guided head position found in this study were similar to the ones found by Billiaert et al.¹⁶ who analysed the reproducibility of the head position in a sitting and standing position. When comparing the two sessions,

the standard deviation (SD) found for the mirror-guided head posture on the pitch axis ($SD = 2.4^\circ$) in this study is similar to the one found by Peng and Cooke ($SD = 4.3^\circ$) for a fifteen-year reproducibility of natural head posture.¹⁰

For the mirror-guided head position, a good correlation was observed on the pitch axis since the mirror affects it. However, the mirror-guided position does not seem to influence the roll and yaw axes, which is why the correlation was much lower for these axes. For the self-balanced head position, good correlation was found even for the roll and the yaw axes. This indicates a similar pattern between the two sessions, meaning that the variation in head position may be specific to each participant.

The limitations in the present study include the potential lack of generalizability as all the included participants were healthy young adults and, thus, represent only a small part of the population. However, the use of this system can be interesting for every population including children, the elderly, and patients with a handicap or other pathological conditions since it is easy to use and allows a recording of the variations of head position over a certain period of time.

The reproducibility found in this study is based on a one-week interval, and on two sessions. A longer period between the recordings and multiple sessions would be necessary to evaluate the long-term reproducibility. Finally, different elements influence the head position, such as fatigue and ocular and respiratory factors, and thus the recordings of each participant should be evaluated when these factors are the same for the two sessions.

5 | CONCLUSION

Good-to-excellent correlations were found between the instantaneous analysis of head position and the five-minute head position recordings. However, on the pitch axis, the head position was systematically more elevated in the five-minute analysis than the instantaneous procedure, whereas no difference was observed for the roll and yaw axes.

Most participants have the same pattern of evolution of their head position over a five-minute recording. An elevation of the head and a rotation to the left were observed as general patterns. Only the elevation of the head, however, is corrected by the mirror-guided position.

The pattern of evolution of the head position and its variation are generally specific to each participant with moderate-to-good correlation between the two sessions. Overall, the means of the five-minute recordings were more reproducible than those of the mirror-guided head position. The roll axis had the least amount of variation, followed by the pitch and finally the yaw axes, respectively.

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CONFLICT OF INTEREST

The authors declare no competing interests.

AUTHOR CONTRIBUTIONS

K.B. and M.A., contributed to the data collection and data analysis, G.S.A. and S.K. contributed to the study design and drafting of the manuscript.

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DATA AVAILABILITY STATEMENT

All data included in this study are available upon reasonable request from the corresponding author.

ORCID

Mustafa Al-Yassary  <https://orcid.org/0000-0002-9075-8194>

Kelly Billiaert  <https://orcid.org/0000-0003-3628-2209>

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DISCUSSION

ÉVALUATION DE LA VALIDITE

La première étude a montré qu'il existe une excellente corrélation in vitro ainsi que in vivo entre le dispositif CROM et l'IMU. Nos résultats sur la posture de la tête sont similaires à ceux trouvés précédemment par Beange et al.[16] qui ont comparé le MetaMotionR à une capture de mouvement à 7 caméras (Vicon MX40) sur la colonne vertébrale avec un degré d'excellente similarité.

Le "yaw" est légèrement moins précis que les deux autres axes. Selon nous, cela est principalement dû au fait que le "pitch" et le "roll" sont donnés par deux inclinomètres, tandis que le "yaw" est donné par une boussole sur le dispositif CROM. La boussole a une plus grande échelle (tous les 2° pour la boussole et tous les 1° sur les inclinomètres) et plus difficile à reporter les valeurs

La différence entre les deux systèmes est légèrement plus grande dans le test in vivo. Les deux raisons les plus importantes semblent être que le patient ne peut pas rester complètement stable et que les relevés de données donnés par le dispositif CROM sont plus compliqués à rapporter. Ces raisons dépendent de l'opérateur et du patient et ne sont pas observées avec les capteurs IMU.

Après avoir utilisé les deux systèmes, les auteurs estiment que les IMU est plus facile à utiliser car ils permettent de mesurer la posture de la tête sur les 3 axes simultanément et ne dépendent pas des relevés d'un opérateur, éliminant ainsi un biais à cet égard. De plus, les capteurs sont très légers et peu encombrants, contrairement au dispositif CROM, ce qui permet de ne pas influencer les mesures de la position naturelle du patient par l'équipement porté.

Cependant, contrairement au CROM qui peut être directement utilisé sur la tête du patient, le système IMU doit être calibré avant la mesure afin de donner des résultats précis.

ÉVALUATION DE LA REPRODUCTIBILITE

Pour mesurer la reproductibilité nous avons pris groupe de 30 jeunes adultes en bonne santé, on leur a demandé de regarder quatre emplacements différents tout en restant debout (en orthoposition)[17] dans une pièce bien éclairée. Cela comprenait un miroir situé à trois mètres devant eux, où on leur a demandé de se regarder, ainsi que trois points différents indiqués par un numéro et un point (d'un diamètre de 0,5 cm). La posture de la tête a été enregistrée à l'aide de l'IMU (unité de mesure inertielle) pendant que les participants regardaient chacun de ces quatre emplacements. La calibration de l'IMU a été effectuée avec les participants regardant leurs yeux dans le miroir (posture de la tête guidée par le miroir), avec un capteur sur le front et l'autre sur le sol utilisé comme référence. Après cette calibration, on a demandé aux participants de regarder chacun des trois points dans l'ordre, en regardant dans le miroir entre chacune des trois positions. Cela a été fait deux fois pour chaque position au cours de la même session. La même procédure a été réalisée à deux reprises, avec un intervalle d'une semaine. Au total, 20 enregistrements ont été effectués pour chaque participant (10 dans chaque session, soit deux pour chaque position et quatre pour le miroir).

Lorsqu'on regarde un point (position de chasse), trois paramètres principaux peuvent influencer la posture de la tête, à savoir la posture du corps[18], le regard des yeux[19] et la position dans le champ de vision[20]. Lorsqu'on demande au participant de maintenir une posture corporelle droite, de ne pas bouger les yeux et de garder le point au centre du champ de vision, il est probable que de petits mouvements de la tête et des yeux puissent quand même se produire, ce qui peut expliquer les variations entre les deux sessions.

Cette variation entre les sessions est toutefois intéressante à étudier. Il a été observé que les participants ayant une faible corrélation lors de la première session avaient également une faible corrélation lors de la deuxième session et vice versa. Cela permet de supposer que la stabilité de la posture de la tête est très spécifique à chaque individu. De plus, il a été remarqué que, dans un groupe de patients en bonne santé, la variation est plus importante avec la rotation et le pitch, mais le roll est très stable.

En ce qui concerne la posture guidée par miroir, nos résultats sont très similaires aux études précédentes concernant le pitch. Solow et Tallgren[12] ont constaté que la NHP, qui se réfère à la posture guidée par miroir de la tête, est très reproductible avec une amplitude de 1,4°. C'est très proche de nos résultats (avec une amplitude de 2,3°). Nous avons également remarqué qu'il y a des variations détectées pour les axes de roll et de yaw.

Si l'on prend en considération les différents facteurs pouvant influencer la posture de la tête pendant la position de chasse, nous pouvons affirmer que ce système IMU présente une bonne reproductibilité compatible pour mener des recherches avancées dans le domaine.

Ces résultats doivent être interprétés avec prudence car la population de notre étude ne concerne uniquement de jeunes adultes en bonne santé, avec une analyse de la NHP en position debout. Il serait intéressant d'évaluer la posture de la tête dans une population plus importante et dans différentes positions (comme la position assise, par exemple).

COMPARAISON ENTRE ANALYSE INSTANTANEE ET ANALYSE SUR 5 MINUTES

Pour la mesure sur 5 minutes, Les participants ont été invités à s'asseoir sur une chaise en position 90-90-90 (correspondant à un angle de 90 degrés au niveau des hanches, des genoux et des pieds) face à un mur blanc situé à un mètre devant eux. On leur a demandé de regarder devant eux et de maintenir une posture droite de la tête en conservant leur sensation naturelle d'équilibre (position d'équilibre naturel). Cette position a été enregistrée pendant 5 minutes. Ensuite, un miroir a été placé sur le mur devant les participants, et on leur a demandé de regarder leurs yeux pendant 20 secondes (position auto-guidée). Cette procédure a été répétée deux fois avec un intervalle d'une semaine.

La présente étude a montré une excellente corrélation entre la procédure instantanée et la moyenne des enregistrements de cinq minutes pour les axes de pitch et de roll, et une bonne corrélation pour l'axe de yaw. L'analyse sur cinq minutes a montré une élévation systématique de la tête par rapport à la procédure instantanée, mais aucun changement systématique n'a été observé pour les axes de roll et de yaw.

Les données enregistrées à la 15ème seconde avec le système IMU ont été utilisées comme procédure instantanée représentative. Ce moment a été choisi car il représente approximativement le temps nécessaire pour prendre une photographie. De plus, contrairement à la photographie, le système IMU permet d'obtenir des données pour les trois axes.

ANALYSE DE LA POSTURE SUR 5 MINUTES

Lorsque les différents segments ont été comparés, il a été constaté que les participants avaient tendance à élever leur tête et à tourner progressivement vers la gauche. Cependant, aucune tendance générale n'a été observée pour l'axe de roll. Lorsque le miroir était affiché, il a été observé que les participants corrigeaient leur élévation sur l'axe de pitch, sans aucun effet sur les deux autres axes (roll et yaw). Les participants qui avaient tendance à tourner leur tête conservaient leur rotation et continuaient dans la même direction.

La variation de la position de la tête semble être influencée par différents facteurs. La rotation de la tête peut être liée à l'œil dominant, qui est souvent l'œil droit. Cette observation est décrite par Pradham et al. qui ont trouvé une forte relation entre la dominance oculaire et la direction des rotations de la tête[21]. Il a également été observé qu'une élévation progressive dans la position de la tête auto-équilibrée se produit au fil du temps jusqu'à un maximum de $1,5^\circ$ après 5 minutes, probablement causée par la fatigue musculaire. Cette élévation de la tête dans la position auto-équilibrée par rapport à la position de la tête guidée par miroir a également été observée par Jakobsone et al.[7]

ANALYSE DE LA REPRODUCTIBILITE D'UNE ANALYSE SUR 5 MINUTES

En comparant les deux sessions, une corrélation plus forte a été trouvée pour la position de la tête auto-équilibrée que pour la position de la tête guidée par miroir sur chaque axe. Les corrélations de la position de la tête guidée par miroir trouvées dans cette étude étaient similaires à celles trouvées par Billiaert et al.[22] qui ont analysé la reproductibilité de la position de la tête en position assise et debout. Lors de la comparaison des deux sessions, l'écart type (SD) trouvé pour la posture de la tête guidée par miroir sur l'axe de pitch ($SD = 2,4^\circ$) dans cette étude est similaire à celui trouvé par Peng et Cooke ($SD = 4,3^\circ$) pour une reproductibilité de quinze ans de la posture naturelle de la tête[23].

Pour la position de la tête guidée par miroir, une bonne corrélation a été observée sur l'axe de pitch car le miroir l'influence. Cependant, la position guidée par miroir ne semble pas influencer les axes de roll et de yaw, ce qui explique la corrélation beaucoup plus faible pour ces axes. Pour la position de la tête auto-équilibrée, une bonne corrélation a été trouvée même pour les axes de roll et de yaw. Cela indique un motif similaire entre les deux sessions, ce qui signifie que la variation de la position de la tête peut être spécifique à chaque participant.

Les limitations de la présente étude comprennent un éventuel manque de généralisabilité car tous les participants inclus étaient de jeunes adultes en bonne santé, représentant ainsi seulement une petite partie de la population. Cependant, l'utilisation de ce système peut être intéressante pour toutes les populations, y compris les enfants, les personnes âgées et les patients handicapés ou atteints d'autres conditions pathologiques, car il est facile à utiliser et permet d'enregistrer les variations de la position de la tête sur une certaine période. La reproductibilité trouvée dans cette étude est basée sur un intervalle d'une semaine et sur deux sessions. Une période plus longue entre les enregistrements et plusieurs sessions seraient nécessaires pour évaluer la reproductibilité à long terme. Enfin, différents éléments influencent la position de la tête, tels que la fatigue et les facteurs oculaires et respiratoires, et donc les enregistrements de chaque participant doivent être évalués lorsque ces facteurs sont les mêmes pour les deux sessions.

FUTURE RECHERCHE DANS LE DOMAINE

À l'issue de nos recherches, nous avons mis en place un protocole efficace permettant de mesurer avec précision la posture de la tête. Nous pensons que ce protocole pourrait avoir des implications significatives pour les futures études dans ce domaine. Tout d'abord, cette méthode nous permettra de mieux comprendre l'influence de la posture sur la croissance maxillo-faciale. Nous pourrions ainsi déterminer si une posture anormale est une cause, une conséquence ou simplement associée aux malocclusions.

En outre, nous pourrions approfondir notre compréhension de l'impact de chaque axe sur la croissance maxillo-faciale et analyser les conséquences spécifiques liées à chacun d'eux. Cela nous aidera à évaluer ce qui est considéré comme une posture normale et pathologique, en définissant clairement les degrés de déviation qui sont considérés comme acceptables et ceux qui pourraient entraîner des conséquences néfastes. Une autre question cruciale que cette approche nous permettra d'explorer est de savoir si une correction de la posture peut prévenir le développement de malocclusions. De même, nous pourrions étudier si la correction des malocclusions peut, dans un sens inverse, contribuer à corriger la posture. Ces aspects sont essentiels pour établir des stratégies de traitement efficaces et mieux comprendre les liens complexes entre la posture de la tête et les problèmes orthodontiques.

Nous avons une étude en cours qui se penche sur la correction de l'occlusion croisée postérieure unilatérale fonctionnelle chez les enfants. L'hypothèse que nous cherchons à vérifier repose sur l'idée que ces patients pourraient présenter une posture de la tête potentiellement anormale,

pouvant contribuer à l'apparition de la crossbite. Notre approche vise à comparer la posture de la tête entre un groupe témoin et les patients atteints de crossbite unilatérale fonctionnelle, en utilisant les IMU. Si nos résultats confirment que la posture de la tête est effectivement différente entre ces deux groupes, cela ouvrirait des perspectives quant à la correction de la malocclusion. En effet, nous nous interrogeons sur la possibilité que la correction de l'occlusion puisse également influencer la posture de la tête, suggérant ainsi une relation complexe entre l'anomalie occlusale et la biomécanique de la posture.

CONCLUSIONS

Les capteurs IMU présentent une excellente corrélation avec le « cervical angle of motion » (CROM), tant pour les tests in vitro que in vivo. Lorsqu'ils sont correctement calibrés, les capteurs IMU peuvent être une méthode fiable pour analyser la posture de la tête. Les variations de la NHP sont d'environ 2° pour le pitch et légèrement moins pour les axes de roll et de yaw. La stabilité de la posture de la tête est spécifique à chaque personne et varie en fonction de l'axe.

Des corrélations allant de bonnes à excellentes ont été trouvées entre l'analyse instantanée de la position de la tête et les enregistrements sur cinq minutes. Cependant, sur l'axe de pitch, la position de la tête était systématiquement plus élevée dans l'analyse sur cinq minutes que dans la procédure instantanée, tandis qu'aucune différence n'a été observée pour le roll et le yaw.

La plupart des participants présentent le même schéma d'évolution de leur position de tête sur un enregistrement de cinq minutes. Une élévation de la tête et une rotation vers la gauche ont été observées comme modèles généraux. Cependant, seule l'élévation de la tête est corrigée par la position guidée par le miroir.

Le schéma d'évolution de la position de la tête et sa variation sont généralement spécifiques à chaque participant, avec une corrélation modérée à bonne entre les deux sessions. Dans l'ensemble, les moyennes des enregistrements sur cinq minutes étaient plus reproductibles que celles de la position de la tête guidée par le miroir. L'axe de roll présentait le moins de variation, suivi par le pitch et enfin les axes de yaw, respectivement.

En conclusion, notre protocole ouvre de nouvelles perspectives dans le domaine de l'orthodontie en permettant une analyse approfondie de la posture de la tête. Les découvertes potentielles qui découleront de cette méthode nous aideront à mieux cerner les mécanismes de la croissance maxillo-faciale et à établir des approches thérapeutiques plus personnalisées et efficaces pour les malocclusions.

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